

Awad

Sr. Python Data Engineer

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Green Card



Professional Summary:

- Over 9+ years of experience as a Senior Python Data Engineer, specializing in cloud computing, data engineering, and AI-based automation across finance, manufacturing, and airline domains.
- Expert in Python, PySpark, SQL, and ETL pipeline development for building scalable, high-performance data solutions.
- Strong proficiency in Azure, AWS, and GCP for designing, deploying, and maintaining cloud-native data and AI architectures.
- Skilled in Kubernetes, Docker, Terraform, and Ansible for container orchestration and Infrastructure as Code (IaC) automation.
- Experienced in designing and managing end-to-end data pipelines using Azure Data Factory, Snowflake, Informatica, and Databricks.
- Developed and optimized machine learning and NLP pipelines using TensorFlow, PyTorch, and OpenAI frameworks for intelligent automation.
- Hands-on experience in CI/CD automation with Azure DevOps, Jenkins, GitHub Actions, and ArgoCD (GitOps).
- Proficient in data modeling, cleansing, and transformation using advanced Python libraries — Pandas, NumPy, and SciPy.
- Deep understanding of real-time data streaming with Kafka, Spark Streaming, and Event Hubs for high-throughput systems.
- Integrated observability and monitoring frameworks using Splunk, Grafana, and Prometheus to ensure reliability and performance.
- Strong background in Linux system administration, network configuration, and security best practices for containerized environments.
- Collaborated with data scientists and AI teams to operationalize ML models into production-ready systems.
- Adept at building RESTful APIs, backend services, and data automation frameworks using Django, Flask, and FastAPI.
- Recognized for mentoring teams, driving process efficiency, and implementing best practices in data infrastructure management.
- Passionate about cloud transformation, AI integration, and automation-first engineering, with a focus on performance, scalability, and innovation.

Technical Skills:

Programming & Scripting	Python (Advanced), PySpark, Golang, Java, Shell, Bash, PowerShell, SQL, PL/SQL
Frameworks & Libraries	Django, Flask, FastAPI, Django REST Framework, Pandas, NumPy, SciPy, Matplotlib, TensorFlow, PyTorch, Keras, OpenAI, Scikit-learn, LangChain
Cloud Platforms	Microsoft Azure, Amazon Web Services (AWS), Google Cloud Platform (GCP), OpenStack
Databases	MySQL, PostgreSQL, SQL Server, Oracle, MongoDB, Cassandra, CouchDB, Redshift, DynamoDB
Monitoring Tool	Zabbix, Splunk, Nagios, Dynatrace, Blue stripe
Web & API Development	RESTful APIs, Microservices, JSON, SOAP, GraphQL, HTML5, CSS3, JavaScript, Angular, React (basic)
Machine Learning & AI	NLP (BERT, LSTM), Generative AI, Model Deployment (SageMaker, MLflow), Data Science Pipelines, Semantic Search
Operating Systems	Linux (RedHat, CentOS, Ubuntu, SUSE), Windows Server, Solaris, AIX, HP-UX
Business Intelligence & Visualization	Power BI, Tableau, Python Seaborn, Plotly, Matplotlib

Professional Education Experience:

Apple INC
Cupertino,CA
Senior Python Data Engineer

June 25 - Present

Project: Data Platform & AI Enablement

Responsibilities:

- Took ownership of core Python and PySpark data pipelines supporting large-scale analytics and machine learning workloads across multiple Apple data platforms.
- Designed and optimized batch and streaming pipelines to process high-volume datasets used by product analytics and operational teams.
- Partnered with data scientists and platform engineers to productionize ML models, focusing on data preparation, feature pipelines, and deployment readiness.
- Built and maintained REST APIs using FastAPI and Flask to expose curated datasets and model outputs to internal consumers.
- Improved pipeline reliability by implementing data validation, monitoring, and alerting, reducing data quality issues in downstream systems.
- Worked closely with DevOps teams to deploy containerized workloads using Docker and Kubernetes, following internal CI/CD standards.
- Optimized query performance and storage layouts in cloud data warehouses, improving reporting and analytics response times.

- Actively supported production environments, troubleshooting data issues and participating in on-call rotations.

Environment: Python, PySpark, FastAPI, Flask, Kafka, Spark, Databricks, Snowflake, Docker, Kubernetes, Git, CI/CD, Linux

Northern Trust
Chicago, IL
Sr Python Data Engineer

June 22 – May 25

Project: Digital Asset Innovation Platform

Responsibilities:

- Led the design and development of scalable Python and PySpark data pipelines supporting high-volume financial and digital asset datasets used by analytics and reporting teams.
- Built and supported real-time ingestion workflows using Kafka, Spark Streaming, and Azure Event Hubs to enable near-real-time portfolio and market insights.
- Partnered closely with data scientists to productionize machine learning models, handling feature engineering, data validation, and automated retraining workflows.
- Developed REST APIs using Flask and Django to expose curated datasets and model outputs to downstream applications and internal consumers.
- Implemented data quality checks, reconciliation logic, and monitoring to improve pipeline reliability and reduce downstream data issues.
- Deployed and managed ML workloads using Azure Machine Learning and AWS SageMaker, ensuring consistent promotion across environments.
- CI/CD pipelines through Azure DevOps, enabling automated testing and deployment of data and ML components.
- Designed governance and lineage solutions using Azure Purview and Snowflake metadata to improve auditability and compliance.
- Built Power BI and Grafana dashboards to surface KPIs, risk indicators, and operational metrics for business stakeholders.
- Actively supported production systems, troubleshooting performance issues and optimizing cost and resource usage across cloud services.

Environment: Python, PySpark, Azure Data Factory, Databricks, Kafka, Snowflake, Flask, Django, Azure ML, AWS SageMaker, Power BI, Grafana, Splunk, Git, Linux

Caterpillar Inc,
Irving, TX
Python Data Engineer

May 20 – June 22

Project: Intelligent Asset Data Platform

Responsibilities:

- Developed and optimized batch and streaming data pipelines to process large volumes of telemetry data from connected industrial equipment.
- Built ETL workflows using AWS Glue, S3, Snowflake, and Azure Databricks to consolidate IoT data into analytical datasets used by engineering and operations teams.

- Supported predictive maintenance initiatives by preparing training data and integrating ML models developed in TensorFlow and Scikit-learn.
- Implemented real-time data streams using Kafka, Azure Event Hubs, and AWS Kinesis to enable continuous equipment monitoring.
- Containerized data processing and inference services using Docker and Kubernetes (AKS) to support scalable deployments.
- Developed API services with Flask and FastAPI for anomaly detection, sensor data access, and downstream integrations.
- Introduced MLflow-based model versioning and monitoring to improve visibility into model performance over time.
- Collaborated with cross-functional teams to integrate C3 AI capabilities with cloud-native data services.
- Built operational dashboards using Grafana and Prometheus, improving visibility into pipeline health and system performance.

Environment: Python, PySpark, Azure Databricks, Kafka, AWS Glue, Snowflake, Docker, Kubernetes, MLflow, FastAPI, Grafana, Prometheus, Terraform.

American Express
Phoenix, AZ
Sr. Python Developer

Oct 17 – April 20

Project: Payment Risk & Data Intelligence Platform

Responsibilities:

- Designed and maintained Python-based ETL and orchestration workflows supporting fraud detection and payment analytics systems.
- Built streaming pipelines using Kafka and Spark to process high-volume transaction data for near-real-time risk scoring.
- Developed and maintained feature pipelines feeding machine learning models used in fraud prediction and anomaly detection.
- Deployed ML workflows on AWS SageMaker and integrated inference results into REST APIs consumed by internal systems.
- Created data ingestion and validation services using Flask and FastAPI, improving data consistency across platforms.
- Implemented automated data quality checks and reconciliation processes to reduce false positives in fraud workflows.
- Built reusable feature stores in Databricks and Snowflake to support multiple modeling initiatives.
- Worked closely with DevOps teams to support CI/CD pipelines and containerized deployments.

Environment: Python, PySpark, Airflow, Kafka, Spark, AWS Glue, SageMaker, Databricks, Snowflake, Flask, FastAPI, Docker, Git, Linux

**Southwest Airlines,
Dallas, TX
Python Developer**

Aug 2016 – Sep 2017

Project: Passenger Analytics & Operational Data Hub

Responsibilities:

- Developed **Python-based ETL workflows** to unify booking, flight, and customer datasets into a centralized data warehouse.
- Developed Python-based ETL workflows to integrate booking, flight, and customer data into a centralized analytics platform.
- Built cloud-based pipelines using Azure Data Factory and AWS Glue to populate Snowflake and Redshift warehouses.
- Supported analytics use cases including passenger segmentation, demand forecasting, and operational reporting.
- Developed Flask-based APIs to deliver curated passenger and flight data to internal applications.
- Assisted in migrating legacy, on-premise ETL processes to cloud infrastructure using Docker and AWS services.
- Created Power BI dashboards and Python visualizations to support operational and planning teams.
- Automated deployment and infrastructure provisioning using Terraform and Ansible.

Environment: Python, SQL, Azure Data Factory, AWS Glue, Snowflake, Redshift, Flask, Power BI, Docker, Terraform, Ansible, Linux

Educational Details:

Master's in computer information and technology, **California University of management and science -2015**